Second Moment of Degree Three L-functions

Thesis Defence Seminar

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L-functions

Let $L(f,s)$ be an L-function associated to an automorphic form f represented by a Dirichlet series

$$L(f,s) = \sum_{n=1}^{\infty} a_f(n) n^{-s}, \quad \text{for} \quad \Re(s) > 1.$$

Definition 1

$2k$th moment of $L(f,s)$ on the critical line:

$$M_f^k(T) := \int_0^T \left| L\left(f, \frac{1}{2} + it\right) \right|^{2k} dt. \tag{1}$$

Conjecture:

- Generalized Lindelof Hypothesis:

$$L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^\varepsilon \implies M_f^k(T) \ll_{\varepsilon,k} T^{1+\varepsilon}$$

History: Riemann ζ function

Riemann zeta function: $\zeta(s) = \sum_{n=1}^{\infty} n^{-s}$, for $\Re(s) > 1$.

- 2nd moment: Hardy-Littlewood (1916), Atkinson (1949), Balasubramanian (1978)

$$\int_0^T |\zeta(1/2 + it)|^2 dt \sim T \log T$$

- 4th moment: Ingham(1927), Heath-Brown (1977), Motohashi

$$\int_0^T |\zeta(1/2 + it)|^4 dt \sim \frac{T}{2\pi^2} \log T$$

- 12th moment: Heath-Brown(1978)

$$\int_0^T |\zeta(1/2 + it)|^{12} dt \ll_\varepsilon T^{2+\varepsilon}.$$

$GL(2)$ L-functions

- Let $E(z,w)$ be the Eisenstein series and let $L(E(.,w),s)$ be the associated degree-two L-function.

$$L(E(\cdot,1/2),s) = \zeta^2(s) = \sum_{n=1}^{\infty} d(n)n^{-s}, \quad \text{for} \quad \Re(s) > 1.$$

- The second moment of $L(E(\cdot,1/2),s)$ is actually the fourth moment of the Riemann ζ function.
- $E(z,w)$ is a non-cuspidal automorphic form for $SL(2,\mathbb{Z})$.
- What is the cuspidal analogue for

$$\int_T^{2T} |\zeta(1/2+it)|^4 dt = \int_T^{2T} |L(E(\cdot,1/2),1/2+it)|^2 dt \quad ?$$

$GL(2)$ L-functions

- Let f be a holomorphic/Maaß cusp form for $SL(2,\mathbb{Z})$ and $L(f,s)$ be the associated $GL(2)$ L-function.
- Thus
$$\int_T^{2T} |L(f,1/2+it)|^2 dt$$
would be a cuspidal analogue of the fourth moment of the Riemann zeta function.
- Unfortunately, the Fourier coefficients of f do not share the nice arithmetic properties of $d(n)$. Hence, the method used by Ingham to obtain the fourth moment of the Riemann zeta function does not generalize to the cuspidal case.
- For f, a holomoprhic cusp form for $SL(2,\mathbb{Z})$, Good(1982) derived the first asymptotic expression
$$\int_0^T |L(f,1/2+it)|^2 dt \sim cT \log T + O(T^{2/3+\varepsilon}).$$

$GL(3)$ L-functions

For $\nu = (\nu_1, \nu_2) \in \mathbb{C}^2$, let $E_{P_{\min}}(z, \nu)$ be the minimal parabolic Eisenstein series for $SL(3, \mathbb{Z})$. For $\nu = (1/3, 1/3)$, $L(E_{P_{\min}}(\cdot, \nu), s) = \zeta^3(s)$

$$M^1_{E_{P_{\min}}(\cdot,\nu)} = \int_0^T |L(E_{P_{\min}}(\cdot, \nu), 1/2 + it)|^2 dt = \int_0^T |\zeta(1/2 + it)|^6 dt$$

- Trivial bound: $M^1_{E_{P_{\min}}(\cdot,\nu)} \ll T^{3/2+\varepsilon}$
- Heath-Brown's twelfth moment: $M^1_{E_{P_{\min}}(\cdot,\nu)} \ll T^{5/4+\varepsilon}$

$E_{P_{\min}}(z, \nu)$ is a non-cuspidal automorphic form for $SL(3, \mathbb{Z})$. Therefore, Heath-Brown's twelfth moment implies a non-trivial upper bound for the second moment of $L(E_{P_{\min}}(\cdot, \nu), s)$. What would be the cuspidal analogue?

$GL(3)$ L-functions

Let F be a Hecke-Maaß cusp form for $SL(3,\mathbb{Z})$ and $A(m,n)$ be its normalized Fourier coefficients.

$$L(F,s) = \sum_{n=1}^{\infty} A(1,n)n^{-s}, \quad \Re(s) > 1$$

Remark

For brevity we will denote $A(1,n)$ by $A(n)$.

But the method of Heath-Brown's twelfth moment does not generalize to the second moment for $L(F,s)$. Our main result is the first non-trivial upper bound for the second moment of $L(F,s)$:

Theorem 1 (P.)

$$\int_{T}^{2T} |L(F,1/2+it)|^2 dt \ll_{\varepsilon} T^{3/2-3/32+\varepsilon}.$$

- Previous result:

Theorem 2 (Aggarwal-Leung-Munshi)

$$\int_{T-M}^{T+M} |L(F, 1/2+it)|^2 dt \ll T^{\varepsilon} \left(\frac{T^{9/4}}{M^{3/2}} + \frac{M^3}{T^{21/20}} + M^{7/4} T^{3/40} + M^{15/14} T^{15/28} \right)$$

- Newer result:

Theorem 3 (Dasgupta-Leung-Young)

$$\int_{T}^{2T} |L(F, 1/2+it)|^2 dt \ll_{\varepsilon} T^{4/3+\varepsilon}$$

Subconvexity Bound

Let $L(F, s)$ be an L-function associated to an automorphic form F.

- Analytic conductor:

$$\mathfrak{q}(s, F) = q(F) \prod_{j=1}^{d} (|s + \kappa_j| + 3)$$

- Phragmen-Lindelöf principle $\implies$ Convexity bound:

$$L(1/2 + it, F) \ll \mathfrak{q}(1/2 + it, F)^{1/4+\epsilon}$$

- Genralized Lindelöf Hypothesis predicts:

$$L(1/2 + it, F) \ll \mathfrak{q}(1/2 + it, F)^{\epsilon}$$

- Subconvexity bound:

$$L(1/2 + it, F) \ll \mathfrak{q}(1/2 + it, F)^{1/4-\delta} \quad \text{for } \delta > 0$$

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Subconvexity Bound

Analytic conductor: $\mathfrak{q}(s,F) = q(F)\ \prod_{j=1}^{d}(|s+\kappa_j|+3)$

It is hard to improve the exponent of all the parameters of $\mathfrak{q}(s,F)$ simultaneously, so one often proves a subconvex bound in one parameter or 'aspect' at a time: (keeping the other parameter fixed)

- t-aspect : $q(F)\prod_{j=1}^{d}(|1/2+it+\kappa_j|+3)$
- spectral aspect: $q(F)\prod_{j=1}^{d}(|1/2+it+\kappa_j|+3)$
- level aspect: $q(F)\prod_{j=1}^{d}(|1/2+it+\kappa_j|+3)$

We will primarily focus on the t-aspect and occessionally on the spectral aspect.

Subconvexity Bound for $GL(3)$ L-functions

Let F be a Hecke-Maaß cusp form for $SL(3,\mathbb{Z})$.

- Subconvexity bound (t-aspect):
  $L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{3/4-\delta}$, for $\delta > 0$.

Previous results:

- Munshi: $L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{11/16+\varepsilon}$
- Aggarwal-Leung-Munshi: $L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{5/8+\varepsilon}$

$$\int_T^{2T} |L(F, 1/2 + it)|^2 \, dt \ll_\varepsilon T^{\frac{3}{2}-\delta+\varepsilon}$$

$$\implies L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{\frac{3}{4}-\frac{\delta}{2}+\varepsilon}$$

Corollary 1 (P.)

$$L(F, 1/2 + it) \ll (1 + |t|)^{3/4-3/64+\varepsilon}$$

Subconvextiy Bound for Self-dual $GL(3)$ L-functions

- F be a self-dual Hecke-Maaß cusp form for $SL(3,\mathbb{Z})$
- f_j be a Hecke-Maaß cusp form for $SL(2,\mathbb{Z})$ of Laplacian eigenvalue $\frac{1}{4}+t_j^2$.
- $L(F\otimes f_j,s)$ be the Rankin-Selberg L-function.

Convexity bound:

- t-aspect: $L(F,1/2+it)\ll_\varepsilon (1+|t|)^{3/4+\varepsilon}$
- $GL(2)$ spectral aspect: $L(F\otimes f_j,1/2)\ll_\varepsilon (1+|t_j|)^{3/2+\varepsilon}$

Remark

We will only focus on the t-aspect subconvexity bound as the $GL(2)$ spectral aspect bound would be exactly twice of that.

Subconvextiy Bound for Self-dual $GL(3)$ L-functions

Let us denote the exponent of the subconvex bound (t-aspect) by η:

$$L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{\eta+\varepsilon} \text{ and } L(F \otimes f_j, 1/2) \ll_\varepsilon (1 + |t_j|)^{2\eta+\varepsilon}$$

- Li: $L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{\frac{11}{16}+\varepsilon}$
- Lin-Nunes-Qi: $L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{\frac{3}{5}+\varepsilon}$
- $M_F^1(T) \ll_\varepsilon T^{\frac{3}{2}-\delta+\varepsilon}$: $L(F, 1/2 + it) \ll_\varepsilon (1 + |t|)^{\frac{3-2\delta}{5-2\delta}+\varepsilon}$

Corollary 2 (P.)

$$L(F, 1/2+it) \ll_\varepsilon (1+|t|)^{45/77+\varepsilon} \textit{ and } L(F\otimes f_j, 1/2) \ll_\varepsilon (1+|t_j|)^{90/77+\varepsilon}$$

Rankin Selberg Problem

Let f be a normalized holomorphic Hecke cusp form or a Hecke-Maaß cusp form for $SL(2,\mathbb{Z})$ and let $\lambda_f(n)$ be its n-th Hecke eigenvalue.

$$\Delta(X) = \sum_{n \leq X} \lambda_f(n)^2 - cX, \quad c = \frac{L(Sym^2 f, 1)}{\zeta(2)}$$

- Rankin (1939), Selberg (1940): $\Delta(X) \ll_\varepsilon X^{\frac{3}{5}+\varepsilon}$
- Huang (2021): $\Delta(X) \ll_\varepsilon X^{\frac{3}{5}-\frac{1}{560}+\varepsilon}$
- $M_F^1(T) \ll_\varepsilon T^{\frac{3}{2}-\delta+\varepsilon} \implies \Delta(X) \ll_\varepsilon X^{\frac{3-2\delta}{5-2\delta}+\varepsilon}$

Corollary 3 (P.)

$$\Delta(X) \ll_\varepsilon T^{\frac{45}{77}+\varepsilon} = T^{\frac{3}{5}-\frac{6}{385}+\varepsilon}.$$

Proof of the Second Moment: Splitting the Integral

Goal: $$M_F(T) := \int_{2T}^{3T} |L(F, 1/2 + it)|^2 dt \ll_\varepsilon T^{3/2-3/32+\varepsilon}.$$

- Split the t-integral into shorter ranges with extra average over the range

$$\int_{2T}^{3T} |\cdots|^2 dt \rightsquigarrow \sum_{\frac{T}{\xi} \le r < \frac{2T}{\xi}} \int_{T+r\xi}^{T+(r+1)\xi} |\cdots|^2 dt$$

$$\ll \sum_{\frac{T}{2X} \le r < \frac{2T}{X}} \frac{1}{X} \int_{\xi \sim X} \int_{T+r\xi}^{T+(r+1)\xi} |\cdots|^2 dt \; d\xi.$$

- Approximate functional equation leads to ($H = \frac{N}{X}$, $N = T^{3/2}$)

$$\mathfrak{M} := \int_{\xi \sim X} \sum_{h \ll H} \sum_{n \sim N} A(n)A(n+h) e\left(\frac{(T + r\xi)h}{2\pi n}\right) d\xi$$

Proof of the Second Moment: Delta method

- $\delta(n) = 1$ if $n = 0$ and $\delta(n) = 0$ if $n \neq 0$.
- Separation of Oscillation:

$$\sum_h \sum_n A(n)A(n+h)\cdots = \sum_h \sum_m A(m) \sum_n A(n) \cdots \delta(m-n-h)$$

- Delta method of Duke-Friedlander-Iwaniec:

$$\delta(n) = \int_x \sum_{q \leq Q} \frac{1}{qQ} \sideset{}{^*}\sum_{a \bmod q} g(q,x) e\left(\frac{an}{q}\right) e\left(\frac{xn}{qQ}\right)$$

$$\mathfrak{M} = \frac{1}{Q^2} \int_{\xi \sim X} \int_{x \sim 1} \sum_{q \sim Q} \sideset{}{^*}\sum_{a \bmod q} \times \sum_{h \ll H} e\left(\frac{-ah}{q}\right)$$
$$\times \sum_{n \sim N} A(n) e\left(\frac{-an}{q}\right) \cdots \times \sum_{m \sim N} A(m) e\left(\frac{am}{q}\right) \cdots W\left(\frac{m-n-h}{Q^2}\right)$$

Remark

We choose $Q \sim \sqrt{\frac{N}{T/X}}$. *We need to save* $T^{3/2}$ *in* $\mathfrak{M}$.

Proof of the Second Moment: Dualization

- Voronoi summation formula (m and n-sum)

$$\sum_{m\sim N} A(m)e\left(\frac{am}{q}\right)e(\cdots) \rightsquigarrow \sum_{m\sim\left(\frac{T}{Q}\right)^3} A(m)S(\bar{a},m;q)I(\cdots),$$

  where $I(\cdots)$ is some integral transform.
- Applying the Voronoi summation formula, we save $\frac{X^{5/4}}{T^{7/8}}$ each in the m and the n-sum.
- We apply the Poisson summation formula in the h-sum.
- After careful analysis of the integral transform at this step, we save $\frac{T^{3/2}}{X}$.

Remark

In total, we have saved $\frac{X^{3/2}}{T^{1/4}}$. We need to save $\frac{T^{7/4}}{X^{3/2}}$ more in $\mathfrak{M}$.

Proof of the Second Moment: Duality Principle

After carefully analyzing the integrals and applying A.M.-G.M. inequality we arrive at

$$\mathfrak{M} \ll \frac{1}{T} \int_{\xi \sim X} \sum_{q \sim Q} \sum_{h \sim \frac{\sqrt{X}}{T^{1/4}}} \left| \sum_{m \sim \left(\frac{T}{Q}\right)^3} A(m) S(\bar{a}, m; q) e(\cdots) \right|^2$$

Applying the duality principle and the Ramanujan bound on average

$$\mathfrak{M} \ll \frac{1}{T} \sum_{m \sim \left(\frac{T}{Q}\right)^3} |A(m)|^2 \Delta \ll \frac{T^{2+\varepsilon}}{Q^3} \Delta,$$

where $\quad \Delta = \sup_{\substack{\alpha \\ ||\alpha||_2 = 1}} \sum_{m \sim \left(\frac{T}{Q}\right)^3} \left| \int_{\xi \sim X} \sum_{q \sim Q} \sum_{h \sim \frac{\sqrt{X}}{T^{1/4}}} \alpha(\xi, q, h) S(\bar{a}, m; q) e(\cdots) \right|^2$

Proof of the Second Moment: Poisson on the m-sum

- After opening up the absolute square we apply the Poisson summation formula on the m-sum

$$\sum_{m\sim(T/Q)^3} S(\bar{h}_1, m; q_1)S(\bar{h}_2, m; q_2)e(\cdots) = \frac{T^3}{Q^3 q_1 q_2}\sum_{m\in\mathbb{Z}} \mathfrak{C}\,\mathfrak{J}$$

where

$$\mathfrak{C} = \sum_{\beta \bmod q_1 q_2} S(\bar{h}_1, \beta; q_1)S(\bar{h}_2, \beta; q_2)e_{q_1 q_2}(m\beta),$$

$$\mathfrak{J} = \int_{y\sim 1} e(\cdots\ y^{1/3})e\Big(-\frac{mT^3 y}{Q^3 q_1 q_2}\Big)dy.$$

- In the diagonal case ($m = 0$), we save T. We were required to save $\frac{T^{7/4}}{X^{3/2}}$, so the bound in the diagonal case is non trivial as long as $X \gg \sqrt{T}$.

Proof of the Second Moment: Off-diagonal

In the off-diagonal ($m \neq 0$) case:

- By reciprocity:

$$\mathfrak{C} \asymp q_1 q_2 \; e\left(\frac{q_2 \bar{q}_1}{m h_1} + \frac{q_1 \bar{q}_2}{m h_2}\right)$$

- By stationary phase analysis

$$\mathfrak{J} \asymp \frac{\sqrt{X}}{\sqrt{T}} e\left(2\sqrt{\frac{q_1 q_2}{m}}\left(\Xi_2 - \Xi_1\right)^{3/2}\right) \text{ where }, \Xi_k = \left(\frac{(T + r\xi_k)}{2\pi q_k^2 h_k}\right)^{1/3}$$

- Size of the phase function (analytic oscillation): T/X.
- From the Poisson summation formula, we save $\frac{T^{3/2}}{X^{3/2}}$.

Proof of the Second Moment: Off-diagonal

- Now we have

$$\sum_m \int_{\xi_1} \sum_{q_1} \sum_{h_1} (\alpha_1) \left(\int_{\xi_2} \sum_{q_2} \sum_{h_2} \bar{\alpha}_2 e\left(\frac{q_2 \bar{q}_1}{m h_1} + \frac{q_1 \bar{q}_2}{m h_2} \right) e(\cdots) d\xi_1 d\xi_2 \right)$$

- Apply Cauchy's inequality on the (ξ_1, q_1, h_1)-sum to get rid of α_1.
- Open up the resulting absolute square
- Apply the Poisson summation formula on the q_1 sum. This produces another character sum $\mathcal{C}$ and oscillatory integral. We take the ξ_1 integral along with the oscillatory integral and denote the two variable oscillatory integral as $\mathcal{I}$.

$$\int_{\xi_1} \sum_{q_1 \sim Q} e\left(\frac{q_2 \bar{q}_1}{m h_1} + \frac{q_1 \bar{q}_2}{m h_2} - \frac{q_2' \bar{q}_1}{m h_1} - \frac{q_1 \bar{q}_2'}{m h_2'} \right) e(\cdots) d\xi_1$$

$$= \frac{Q}{m h_1 h_2 h_2'} \sum_{q_1 \in \mathbb{Z}} \mathcal{C}\, \mathcal{I}$$

Proof of the Second Moment: Off-diagonal

- Anlayze the character sum $\mathcal{C}$

$$\mathcal{C} \asymp S(\cdots, \cdots; mh_1) \cdot h_2 h_2' \cdot \delta(\cdots \bmod h_2 h_2').$$

  We can bound $\mathcal{C}$ by $\sqrt{mh_1}$ on average.
- Treat the double integral $\mathcal{I}$ with multivariable stationaty phase analysis. We save $\frac{T}{X}$.
- Overall in this step we save $\frac{T^{3/4}}{X^{1/2}}$. However, due to Cauchy's inequlaity, effective saving would be $\frac{T^{3/8}}{X^{1/4}}$.
- The total saving turns out to be $\frac{T^{15/8}}{X^{7/4}}$. With the choice of $X = \sqrt{T}$, we are at the boundary. So we need to save a little more.
- We still have ξ_2-integral and an oscillatory term from $\mathcal{I}$ (size of the oscillation $\frac{T}{X}$) .

Proof of the Second Moment: Off-diagonal

- Apply Cauchy's inequality, keeping $\int_{\xi_2'} \alpha_2' e(\cdots)$ inside. We have
$$\int_{\xi_2'} \alpha_2' \int_{\xi_2''} \alpha_2'' \int_{\xi_2} e(\cdots)\ d\xi_2\ d\xi_2'\ d\xi_2''.$$
- by repeated integration by parts on the ξ_2 integral, we restrict the range of $|\xi_2' - \xi_2''| \ll \frac{X}{T/X}$. Thus, we save T/X. Since we have applied Cauchy's inequality twice, the effective saving in $M_F(T)$ is $\frac{T^{1/4}}{X^{1/4}}$.
- Hence, the total saving in the off-diagonal (after duality) is $\frac{T^{15/8}}{X^{7/4}} \cdot \frac{T^{1/4}}{X^{1/4}} = \frac{T^{17/8}}{X^2}$. Equate this saving with the diagonal saving of T to optimally choose $X = T^{9/16}$.
- Hence, in $\mathfrak{M}$, we save $X^{3/2} T^{3/4} = T^{3/2+3/32}$ which gives us $M_F(T) \ll T^{3/2-3/32+\varepsilon}$. $\square$

Thank You

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